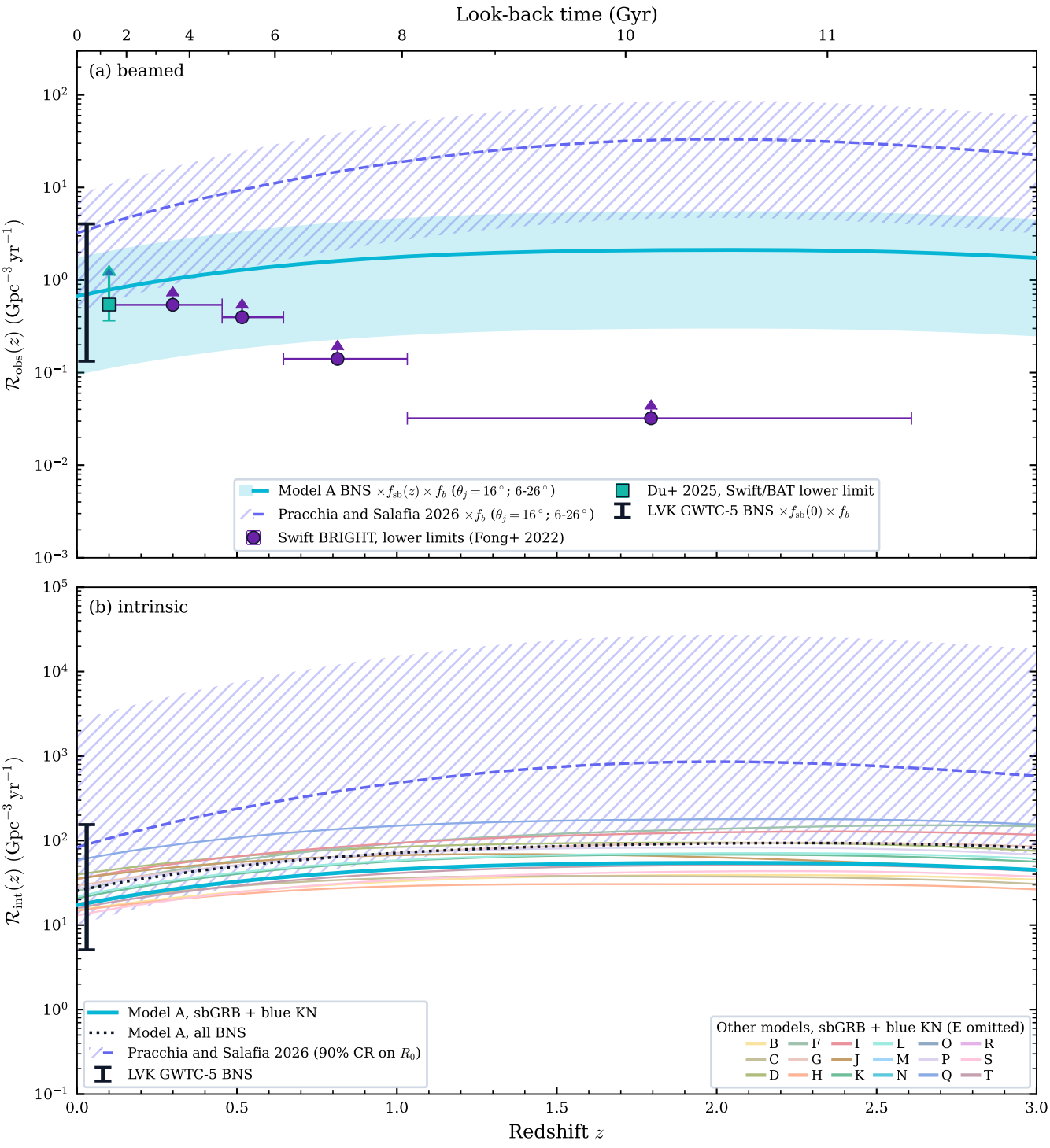

Look-back time (Gyr)
0
2
4
6
8
10
11
(a) beamed
$\mathcal{R}_{\rm obs}(z)$ (Gpc^{-3} yr^{-1})
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
Model A BNS $\times f_{\rm sb}(z) \times f_b$ ($\theta_j = 16^\circ$; 6-26°)
Pracchia and Salafia 2026 $\times f_b$ ($\theta_j = 16^\circ$; 6-26°)
Swift BRIGHT, lower limits (Fong+ 2022)
Du+ 2025, Swift/BAT lower limit
LVK GWTC-5 BNS $\times f_{\rm sb}(0) \times f_b$
(b) intrinsic
$\mathcal{R}_{\rm int}(z)$ (Gpc^{-3} yr^{-1})
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
Model A, sbGRB + blue KN
Model A, all BNS
Pracchia and Salafia 2026 (90% CR on R_0)
LVK GWTC-5 BNS
Other models, sbGRB + blue KN (E omitted)
B F I L O R
C G J M P S
D H K N Q T
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Redshift z